SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Technology Stack

Date	25 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Bootstrap	HTML, CSS, Bootstrap

2	Backend / Server-Side	Python 3.10+, Flask	Lightweight web framework with easy ML model integration; team has strong Python expertise
3	Database / Data Storage	CSV (Crop_recommendation.csv), Pickle (.pkl)	Flat file dataset suitable for ML training; Pickle for efficient model serialization and loading
4	Cloud / Hosting / Deployment	Local Flask Development Server	Cost-effective for demonstration; no cloud dependency; easily deployable on any machine with Python
5	Version Control & CI/CD	GitHub	Seamless team collaboration, version tracking, and code review; free hosting for student projects
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook	Industry-standard ML and data analysis libraries for model training, evaluation, and visualization